PeopleDesk HRMS

Enterprise SaaS Human Resource Management Platform

Technical Architecture Overview

Prepared for Executive Technology Leadership

Document Control

Attribute	Value
Document Title	PeopleDesk HRMS — Technical Architecture Overview
Document Classification	Customer Confidential
Document Status	Released
Version	2.1
Prepared By	PeopleDesk Engineering Team
Prepared For	Enterprise Architecture Review
Generated On	01 June 2026
Last Updated	25 July 2026
Document Owner	Engineering Leadership

Revision History

Version	Date	Author	Description
1.0	01 June 2026	Engineering	Initial Architecture Draft
1.5	10 July 2026	Engineering	Technical Review
2.0	25 July 2026	Engineering	Customer CTO Presentation Release
2.1	25 July 2026	Engineering	Final Release — business value, quality attributes, appendices

Scope & Basis

This document describes the technical architecture of the PeopleDesk HRMS platform as currently implemented and operating in production. It covers solution architecture, technology stack, security architecture, SaaS architecture, data architecture, reporting, deployment, scalability, and integration capability.

All capability statements reflect the deployed implementation. Forward-looking items are presented exclusively in the identified programme subsection (§10) and are not represented as current capability.

Intended Audience

Chief Technology Officer · Chief Architect · Enterprise Architecture Review Board · Solution Evaluation Committee · Engineering Leadership

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1. Executive Summary 📋

KEY TAKEAWAYS

✔ Enterprise-grade, multi-tenant HRMS operating in production across 120+ live customers

✔ Four independently deployable services over a unified 574-table enterprise data platform

✔ Current-generation technology throughout — .NET 9, React 18, SQL Server 2022

✔ Production-proven at 20–41 million rows per core operational table

✔ Nineteen functional domains covering the complete employee lifecycle

Why PeopleDesk HRMS

	Position	Basis
✔	**Enterprise Ready**	19 functional domains · 639 permissioned functions · 40+ configurable approval workflows
✔	**SaaS Native**	Tenant-partitioned data model with a three-level organizational hierarchy
✔	**Production Proven**	120+ live customers sustaining 20–41 million rows per core table
✔	**Cloud Deployable**	Containerized services, orchestration manifests, environment-injected configuration
✔	**Open and Integrable**	1,525 documented REST endpoints plus proven enterprise and device connectivity
✔	**Layered Security Controls**	Identity, authorization, transport, application data and audit

Platform Position

PeopleDesk HRMS is a full-lifecycle Human Resource Management platform delivered as multi-tenant Software as a Service. It supports the complete employee journey — from recruitment and onboarding through attendance, payroll, performance and retirement — for large and mid-market enterprises with complex organizational structures, shift-based workforces, statutory payroll obligations and geographically distributed operations.

Platform Scale

Dimension	Measure
Business domains	19 functional modules
Application programming interface	1,525 endpoints
Application screens	781 routes across 26 frontend modules
Standard reports	~130, served by ~75 dedicated data procedures
Data platform	574 tables · 9,501 columns · 18 schemas
Programmable database objects	325
Permission-controlled functions	639
Automated background processes	46
Customers in production	120+

Operating Characteristics

Core transactional tables sustain between 20 and 41 million rows in production. The platform separates interactive, reporting and batch workloads into distinct services so that high-volume processing — such as a monthly payroll run across tens of thousands of employees — cannot degrade interactive user response.

2. Product Overview 🧩

KEY TAKEAWAYS

✔ Nineteen functional domains unified on one data platform and one permission model

✔ 40+ configurable approval workflows spanning the operational estate

✔ 639 individually permissioned functions with discrete access rights

✔ Complete employee lifecycle coverage from recruitment to retirement

Functional Domains

Domain Group	Modules
Core HR	Employee Management · Organizational Framework · Master & Configuration
Time & Attendance	Attendance & Timesheet · Leave & Movement · Roster & Shift Management
Payroll & Benefits	Payroll & Tax · Allowance & Deduction · Loan · Benefits · Retirement (Provident Fund & Gratuity)
Talent Management	Recruitment & Onboarding · Performance Management · Training & Development · Manpower Analysis
Operations & Governance	Approval Workflow · Asset Management · Expense · Employee Self-Service

Capability Metrics

Measure	Value
Application screens and frontend modules	781 routes · 26 modules
Application programming interface endpoints	1,525
Standard reports	~130
Configurable approval request types	40+
Permission-controlled functions	639 menu nodes
Automated scheduled and continuous processes	46

Governance Depth

Access governance is expressed through 639 individually permissioned functions, each carrying discrete view, create, edit and close rights. Operational governance is expressed through more than forty configurable approval workflow types spanning leave, movement, payroll, asset, expense and employee lifecycle events.

3. Solution Architecture 🏗️

KEY TAKEAWAYS

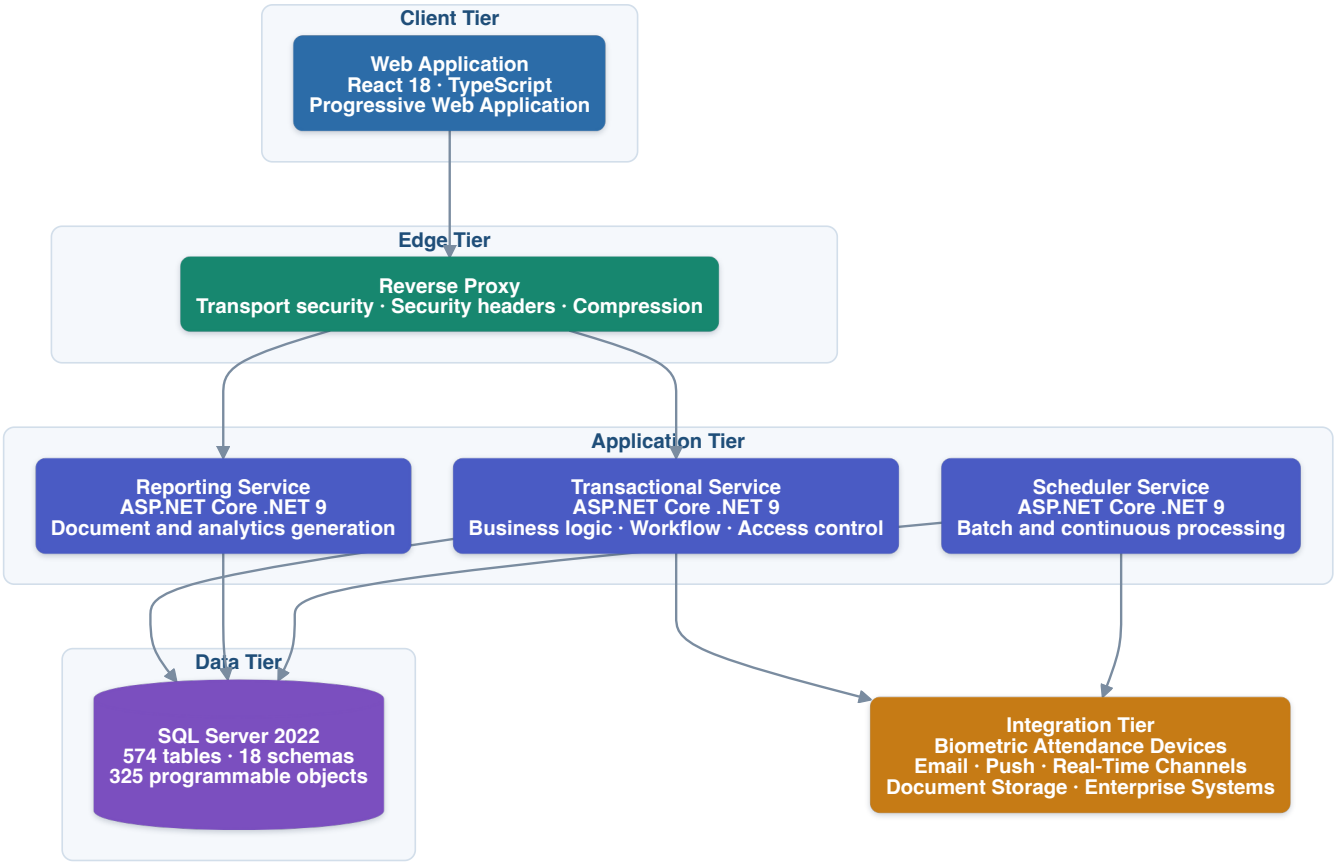
✔ Four independently deployable services, each released on its own cadence

✔ Interactive, reporting and batch workloads physically separated by service

✔ Layered backend across 24 projects with explicit architectural boundaries

✔ Stateless request handling with no server-side session affinity

System Topology



Service Responsibilities

Service	Responsibility
Web Application	Single-page application providing the complete user interface across 781 screens, delivered as an installable progressive web application
Transactional Service	Business logic, workflow orchestration, access control and transactional data operations across 1,525 endpoints
Reporting Service	Document and analytics generation across ~130 standard reports, isolated from transactional workload
Scheduler Service	Batch payroll and attendance processing, leave accrual, attendance device collection, and notification dispatch across 46 automated processes

Architecture Principles

Principle	Application in PeopleDesk HRMS
Separation of Concerns	Interactive, reporting and batch workloads isolated by service boundary
Stateless Services	Token-based identity with no server-side session state
Configuration over Customization	Organizational structure, roles, workflows, policies and document templates configured per tenant
Open API Surface	Platform capability exposed through documented REST interfaces
Layered Security	Controls applied across identity, authorization, transport, application data and audit
Cloud Ready	Containerized services with per-service orchestration manifests
Evidence-Based Evolution	Every change impact-analyzed against a structured knowledge model before implementation

Quality Attributes and Design Response

Quality Attribute	Design Response	Realized By
Scalability	Stateless application services	Token-based identity; no session affinity in the application tier
Availability	Independently deployable services	A reporting or batch failure does not affect interactive availability
Security	Layered security controls	Identity, authorization, transport, application data and audit controls
Maintainability	Modular architecture	24 projects with explicit boundaries; change contained within a boundary
Extensibility	Service separation	New capability added to one service without redeploying the others
Performance	Dedicated reporting service	Document generation never contends with transactional workload
Reliability	Background scheduler isolation	Batch payroll executes outside the interactive request path
Portability	Container-based delivery	One image promoted across environments with injected configuration

Why This Architecture Matters

Architectural Decision	Business Outcome
Independent service deployment	**Faster feature delivery** — a reporting change ships without touching the transactional service
Workload-class separation	**Lower operational risk** — month-end payroll processing cannot degrade the user experience
Stateless application tier	**Enterprise scalability** — capacity is added without session affinity constraints
Modular backend boundaries	**Easier maintenance** — the blast radius of a change is bounded and known in advance
Dedicated reporting tier	**Predictable performance** — heavy report generation is absorbed outside the transactional path
Scheduler isolation	**High-availability foundation** — a batch failure does not remove interactive service
Layered security controls	**Better security isolation** — no single control is the sole barrier
Configuration-led tenancy	**Lower onboarding cost** — a new organization is configured, not coded

Engineering Governance — Knowledge-Driven Development

The platform is developed under a formalized engineering framework in which a twelve-layer structured knowledge base — comprising 162 documents, 32 decision tables and a complete inter-module dependency graph — serves as the authoritative model of the system. Every proposed change is impact-analyzed against that model before implementation proceeds.

The first two phases of this framework are in production use. For an enterprise evaluating a multi-year platform dependency, this addresses a question that architecture alone cannot: whether the engineering organization can evolve the system safely and predictably over time.

4. Technology Stack ⚙️

KEY TAKEAWAYS

✔ Current-generation, long-term-supported technology across every tier

✔ Deliberate dual data-access strategy: ORM for transactional integrity, direct SQL for read throughput

✔ Fully containerized and orchestrated on industry-standard infrastructure

✔ No end-of-life runtime dependency at the platform level

Technology Selection Principles

Platform technology is selected against five criteria, applied consistently across every tier:

Criterion	Meaning
Modern	Current-generation frameworks and runtimes, not legacy branches
Stable	Mature release lines with predictable behaviour under production load
Long-Term Supported	Vendor-supported versions with a defined support horizon
Enterprise Proven	Widely deployed in comparable enterprise environments
Cloud Compatible	Runs unmodified in containerized and orchestrated environments

Platform Technology Matrix

Layer	Technology
Frontend	React 18 · TypeScript · Vite 7 · Redux Toolkit · Ant Design · Material UI · Formik with Yup validation · Progressive Web Application
Backend	ASP.NET Core (.NET 9) · MediatR (CQRS) · FluentValidation · Mapster · SignalR · REST
Data Access	Entity Framework Core 9 for transactional operations · Dapper for high-volume read paths · SQL stored procedures for set-based computation
Database	Microsoft SQL Server 2022
Authentication	JSON Web Token bearer authentication · server-side refresh-token rotation · organizational-scope claims · federated sign-in
Reporting	RDLC (fixed-layout statutory documents) · Handlebars with headless browser rendering (HTML and PDF) · ClosedXML (Excel)
Background Processing	Quartz.NET scheduled jobs · .NET Hosted Services for continuous processing
Infrastructure	Docker multi-stage builds · Reverse proxy · Kubernetes orchestration · Container registry distribution
Build & Release	Azure DevOps pipelines · containerized build and publish · environment-scoped configuration injection
Development Tooling	Git multi-repository workspace · Roslyn static analyzers · documented engineering playbooks · structured knowledge platform

Data Access Strategy

The platform applies a deliberate dual strategy rather than a single access pattern:

Path	Technology	Rationale
Transactional writes	Entity Framework Core 9	Change tracking, audit-column preservation and referential correctness on state-changing operations
High-volume reads	Dapper	Minimal object-mapping overhead on list, search and reporting paths
Bulk computation	SQL stored procedures	Payroll, attendance and accrual logic executed as set-based operations at the data tier

5. Security Architecture 🔒

KEY TAKEAWAYS

✔ Layered security controls across identity, authorization, transport, application data and audit

✔ 639 individually permissioned functions with organizational-scope enforcement

✔ Complete request-level audit trail across all authenticated operations

✔ Controls mapped to the OWASP Secure Coding Practices categories

5.1 Authentication Controls

Control	Implementation
Token-based authentication	JSON Web Token bearer authentication with issuer validation, signing-key validation and zero clock-skew tolerance
Session continuity	Server-side refresh-token issuance and rotation, with token history persisted to support revocation and traceability
Federated identity	Google OAuth sign-in support

5.2 Authorization Controls

Control	Implementation
Granular permission model	639 individually permissioned functions, each carrying discrete view, create, edit and close rights
Role-based assignment	Per-tenant role definitions with permission sets assigned by role
Organizational scope	Business unit, workplace group and workplace scope carried within the security token and applied to data access

5.3 Application Data Protection

Control	Implementation
Application-layer data protection	Request and response payload protection applied between client and service in production environments
Declarative request validation	Validation pipeline evaluates every command against its declared rule set before business logic executes
Centralized exception handling	Structured problem-detail responses; internal failure detail is not returned to clients
Output sanitization	Untrusted markup neutralized across the application interface prior to rendering

5.4 Transport & Edge Controls

Control	Implementation
Transport security	HTTPS enforcement with HTTP Strict Transport Security — one-year maximum age, subdomain inclusion, preload
Frame protection	`X-Frame-Options: DENY`
Content type enforcement	`X-Content-Type-Options: nosniff`
Referrer control	`Referrer-Policy` restricted to same origin
Browser feature restriction	`Permissions-Policy` disabling geolocation, camera, microphone, payment and related device interfaces
Cross-domain policy	`X-Permitted-Cross-Domain-Policies` restricted

5.5 Audit & Traceability Controls

Control	Implementation
Request audit trail	Acting user, route and timestamp recorded for every authenticated operation
Process execution history	Separate success and failure records maintained for every automated background process

5.6 Secure Development Practices

The platform's security controls are mapped to the **OWASP Secure Coding Practices** categories — the industry reference checklist for application-layer security. The table below shows the categories in which controls are implemented in the platform today, and where each is described in this document.

OWASP Category	Implemented Controls	Reference
Input Validation	Server-side validation pipeline evaluates every command against its declared rule set before business logic executes	§5.3
Output Encoding	Untrusted markup neutralized across the application interface prior to rendering	§5.3
Session Management	Token-based sessions with issuer and signing-key validation, zero clock-skew tolerance, and server-side refresh-token rotation with persisted history	§5.1
Access Control	639 individually permissioned functions with role-based assignment and organizational-scope claims carried in the security token	§5.2
Error Handling and Logging	Centralized exception handling with structured problem-detail responses; request-level audit trail and background process execution history	§5.3, §5.5
Communication Security	HTTPS enforcement with HTTP Strict Transport Security, plus the full edge security header set	§5.4
Database Security	Parameterized queries and stored-procedure–based data access across the platform's 235 stored procedures	§7
System Configuration	Current supported framework versions throughout; configuration injected per environment with separated development, staging and production	§4, §9

6. SaaS Architecture ☁️

KEY TAKEAWAYS

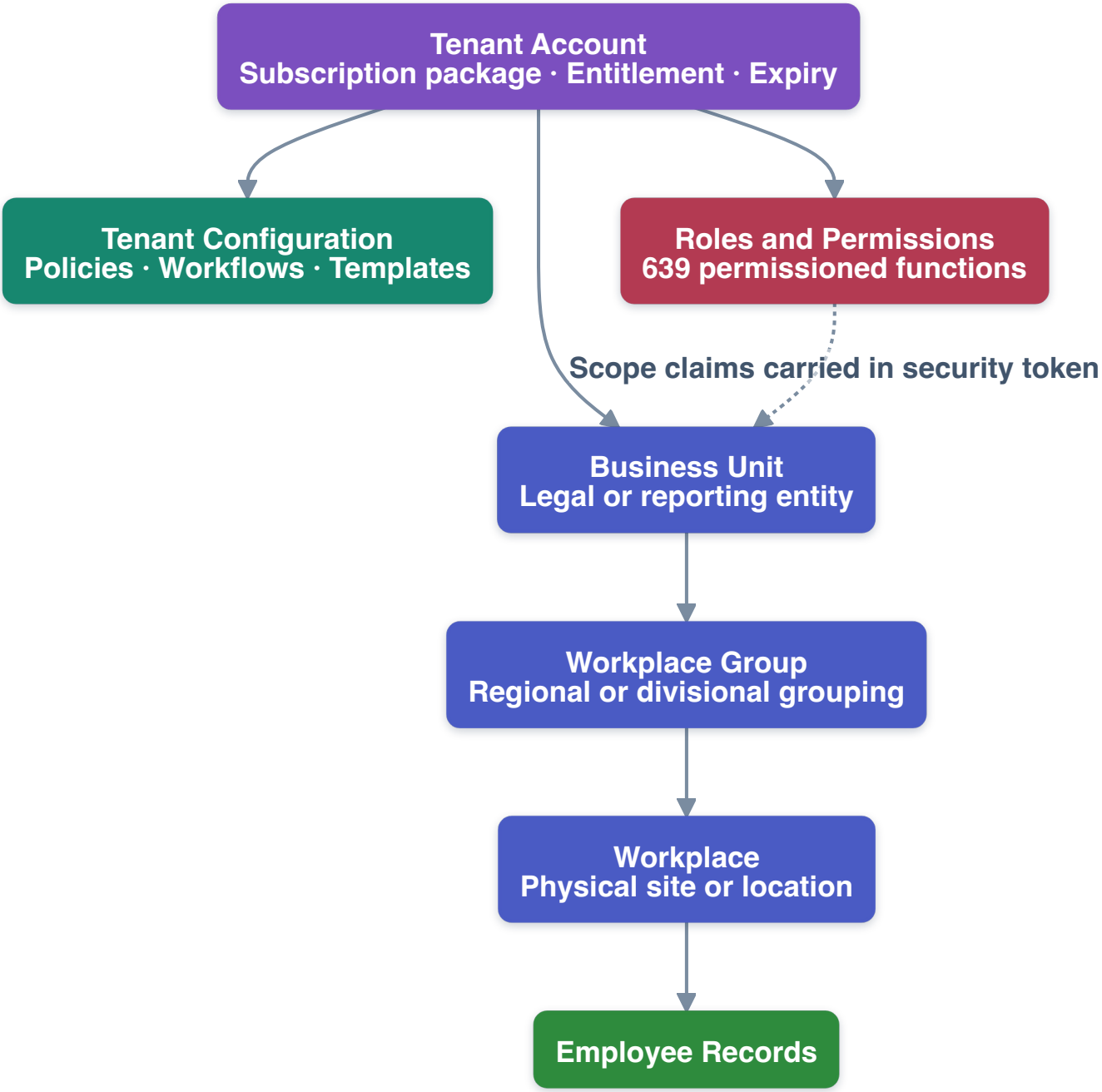
✔ Tenant-partitioned data model applied across the entire data platform

✔ Three-level organizational hierarchy suited to multi-entity, multi-site enterprises

✔ Automated tenant provisioning and subscription entitlement enforcement

✔ Roles, workflows, policies and document templates configured per organization

Tenancy & Organizational Model


Tenant Account
Subscription package · Entitlement · Expiry
Tenant Configuration
Policies · Workflows · Templates
Roles and Permissions
639 permissioned functions
Scope claims carried in security token
Business Unit
Legal or reporting entity
Workplace Group
Regional or divisional grouping
Workplace
Physical site or location
Employee Records

SaaS Capability Matrix

Capability	Implementation
Tenant partitioning	Every business record is keyed to its owning tenant account across the data platform
Organizational hierarchy	Three-level structure — business unit, workplace group, workplace — with scope enforced through security-token claims
Tenant provisioning	Automated account creation flow establishing the organization, seed configuration and initial administrator
Commercial packaging	Three subscription tiers with seat, user and storage entitlements
Entitlement enforcement	Seat limits validated at employee creation; subscription expiry enforced automatically by a daily scheduled process
Per-tenant configuration	Roles, permissions, approval workflows, leave and payroll policies, and document templates configured independently per organization

Subscription Tiers

Tier	Employee Entitlement	User Entitlement	Storage Entitlement
Basic	100	100	1 unit
Lite	500	500	50 units
Premium	50,000	50,000	200 units

Enterprise Relevance

The three-level organizational hierarchy addresses enterprises operating multiple legal entities, regional divisions or physical sites under one corporate structure. Organizational scope is carried in the security token, so a divisional administrator's visibility is constrained by configuration rather than by bespoke query logic.

7. Data Architecture 🗄️

KEY TAKEAWAYS

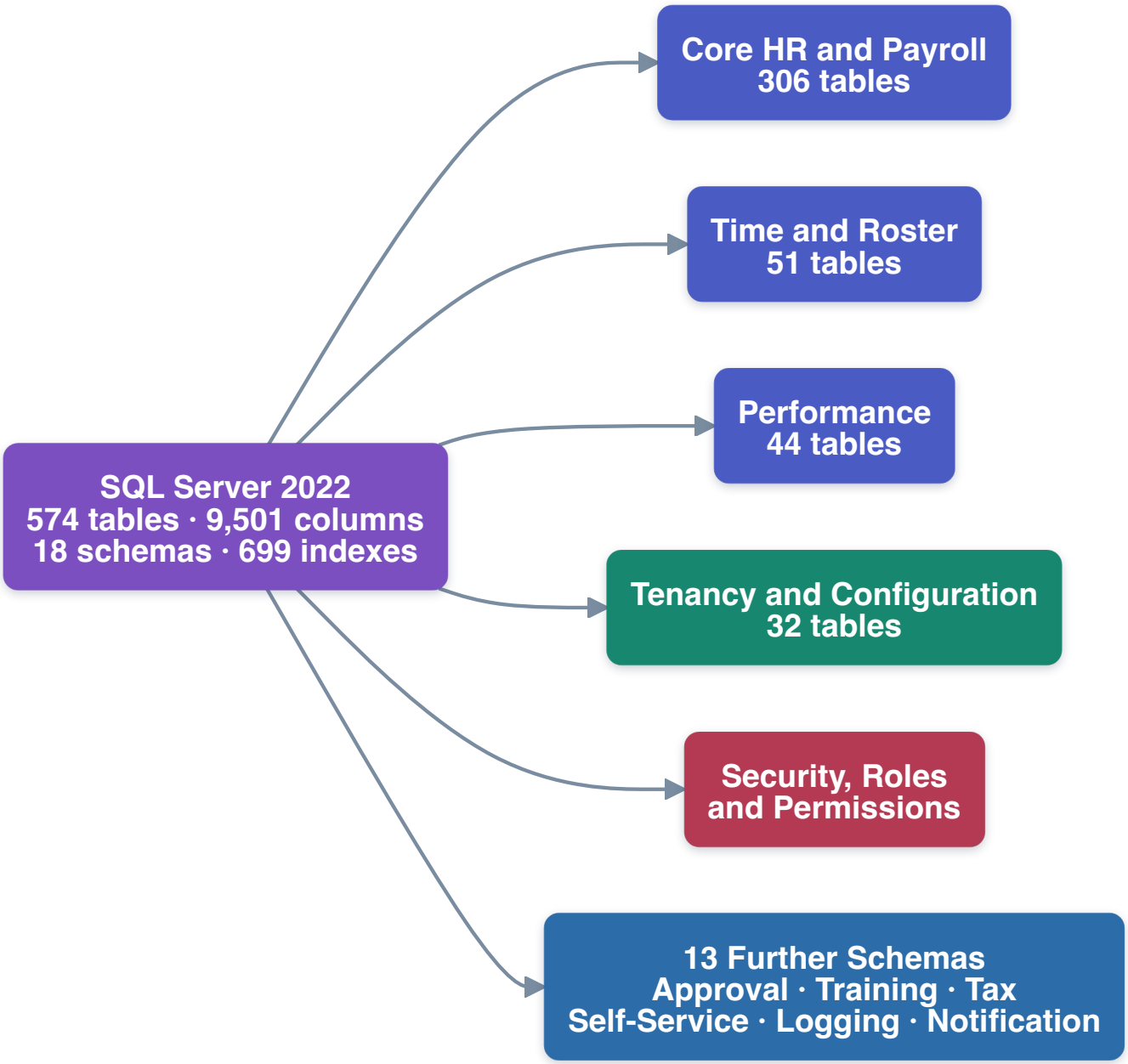
✔ Domain-partitioned enterprise data platform across 18 schemas

✔ Set-based computation executed at the data tier for payroll and attendance at scale

✔ 699 indexes maintained across a 574-table schema

✔ Production-proven at 20–41 million rows per core operational table

Schema Organization



Data Platform Characteristics

Aspect	Approach
Schema organization	Domain-aligned schema separation — security, tenancy, time, performance, approval, tax and logging occupy distinct namespaces rather than a single flat catalogue
Computation strategy	325 programmable objects — 235 stored procedures, 25 table-valued functions, 52 scalar functions and 6 views — execute payroll, attendance and accrual logic as set-based operations at the data tier
Access strategy	Entity Framework Core for transactional writes with change tracking and audit-column preservation; Dapper for high-volume read and reporting paths
Index strategy	699 indexes maintained across the platform
Read efficiency	Cursor-based pagination, no-tracking reads and projection to data transfer objects in the current-generation data access layer

Production Data Volumes

Core operational tables sustain between 20 and 41 million rows in production; the full volume profile is presented in **Section 10 — Scalability & Performance**.

Executing payroll computation as set-based operations at the data tier — rather than through application-tier iteration — is the primary reason the platform sustains these volumes within operational processing windows.

8. Reporting & Analytics 📊

KEY TAKEAWAYS

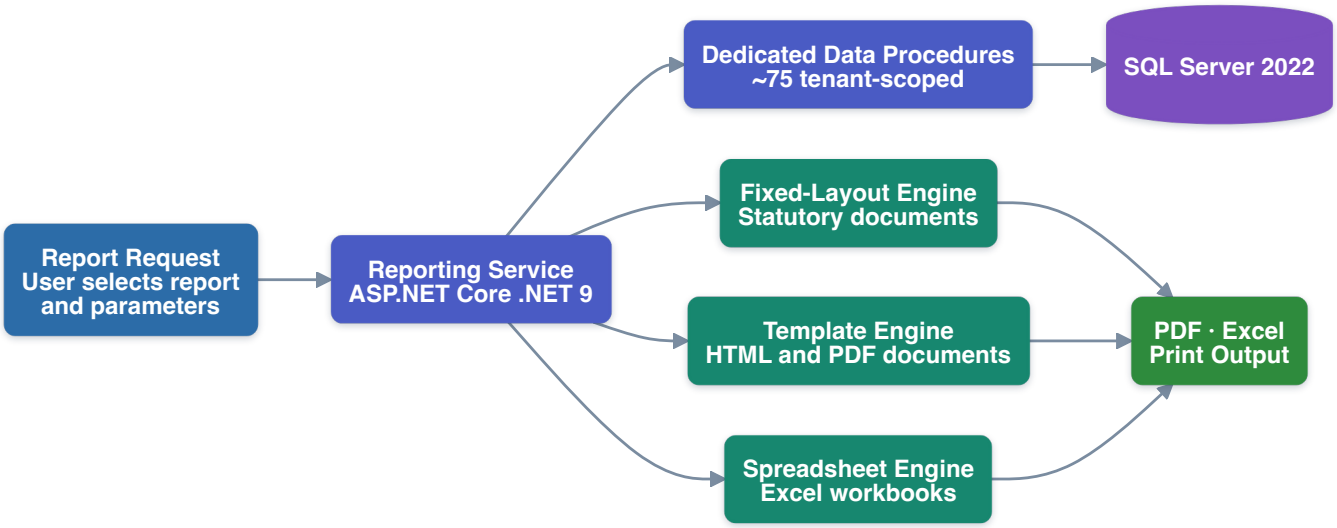
✔ Physically separate reporting service — document generation never contends with transactional workload

✔ ~130 standard reports spanning payroll, statutory, attendance and workforce domains

✔ Render engine selected by output class for fidelity and performance

✔ Organization-specific document templates without forking the report catalogue

Report Generation Flow



Report Coverage

Domain	Reports
Payroll & Statutory	Payslips · salary sheets · bank advice files · bonus and arrear statements · tax certificates · provident fund and gratuity statements · final settlement
Attendance & Time	Attendance registers · overtime · roster · movement · leave balance and history
Workforce & Performance	Manpower analysis · workforce planning · performance appraisal
Operations	Asset · cafeteria and food allowance · letters including increment, promotion and confirmation · identity cards

Rendering Strategy

Output Class	Engine	Rationale
Statutory fixed-layout documents	RDLC	Pixel-exact layout fidelity required for regulatory submission
Designed HTML and PDF documents	Handlebars with headless browser rendering	Template-driven documents supporting organization-specific design
Analytical exports	ClosedXML	Native Excel workbook generation for downstream analysis

Organization-Specific Documents

Customer-specific payslip and salary-sheet templates are supported as data-driven documents within the reporting service. Adding an organization's document format does not require modification to the report catalogue or the rendering engines.

9. Deployment Architecture 🚀

KEY TAKEAWAYS

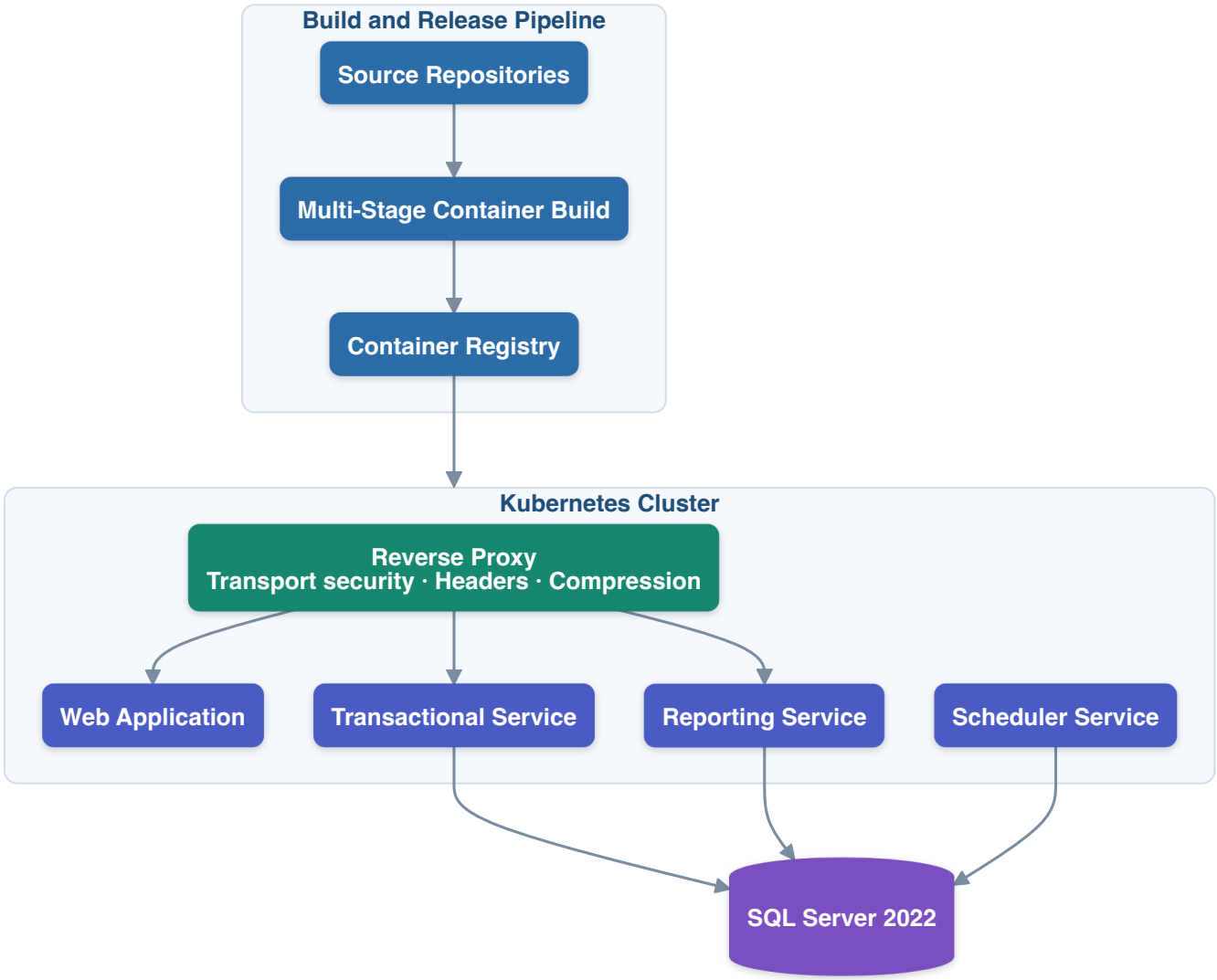
✔ Fully containerized with multi-stage builds across all services

✔ Kubernetes-orchestrated with per-service deployment manifests

✔ One image promoted across environments; configuration injected per environment

✔ Independent release cadence — a reporting change requires no transactional downtime

Deployment Topology



Deployment Characteristics

Aspect	Implementation
Containerization	Multi-stage container builds per service, with build and runtime layers separated; runtime images carry only published output and required native dependencies
Orchestration	Kubernetes deployment and service manifests per service, with registry-based image distribution
Edge layer	Reverse proxy terminating traffic, serving the application shell, applying the security header set and compressing responses
Configuration management	Configuration injected through environment variables per environment; one image serves development, staging and production, differing only by injected configuration
Release model	Each service builds, versions and deploys independently; a reporting release requires no transactional service downtime
Environments	Separate development, staging and production environments with environment-scoped behaviour

Configuration Integrity

Because configuration is injected rather than embedded, the artefact promoted to production is bit-identical to the artefact validated in staging. This removes build-variance as a class of production defect.

> *The deployment topology shown represents the recommended enterprise production architecture. Actual deployment may vary according to customer infrastructure, cloud provider, security policies, and operational requirements.*

10. Scalability & Performance ⚡

KEY TAKEAWAYS

✔ Scalability designed into workload topology, not retrofitted through infrastructure

✔ Batch payroll processing isolated from interactive workload by service boundary

✔ Stateless authentication removes affinity constraint from the application tier

✔ Production-validated at 20–41 million rows per core table

Design Decisions Supporting Scale

Design Decision	Scalability Effect
Workload-class separation	Interactive, reporting and batch processing execute in separate services; batch payroll across tens of thousands of employees does not block interactive users
Stateless authentication	Token-based identity carries no server-side session state, removing affinity constraints from the application tier
Set-based computation	Payroll, attendance and leave accrual execute as data-tier set operations rather than application-tier iteration
Efficient read path	Cursor-based pagination, no-tracking queries and projection to data transfer objects avoid loading entity graphs for read operations
Indexed data platform	699 indexes maintained against a 574-table schema operating at production volume
Frontend delivery optimization	Route-level code splitting across 649 lazy-loaded boundaries, vendor chunk separation, progressive web application asset precaching, and edge compression

Production Load Evidence

Data Class	Rows Sustained
Background process execution history	41.4 million
Raw attendance device capture	27.5 million
Request audit trail	21.4 million
Employee attendance records	20.0 million
Payroll generation detail	6.4 million

Scale-Out Programme

The following workstreams extend the platform's scale-out capacity:

- Distributed caching tier
- Horizontal auto-scaling of the application tier
- Read-replica offload for reporting workload
- Distributed coordination for batch processing

11. Integration Ecosystem 🔌

KEY TAKEAWAYS

✔ Open, standards-based REST interface across 1,525 endpoints

✔ Proven biometric attendance-device integration across 18+ customer environments

✔ Real-time, push and email notification channels operating in production

✔ Enterprise system integration demonstrated with external ERP

Implemented Integration Capability

Category	Capability
Programmatic access	REST interfaces across 1,525 endpoints, secured by token-based authentication and documented through OpenAPI specification
Federated identity	Google OAuth sign-in
Biometric attendance devices	Device-initiated capture through an industry-standard push protocol endpoint, complemented by 18+ dedicated collectors retrieving punch data from customer attendance servers on schedule
Real-time notification	Persistent notification channel delivering live in-application events
Mobile and push notification	Firebase Cloud Messaging delivery
Electronic mail	SMTP delivery for notifications, approval routing and system correspondence
Document storage	Azure Blob Storage and FTP-based document repositories
Enterprise systems	Managerium ERP integration for user and payroll data exchange
Output and hardware	Excel and PDF export across the platform; QR code generation with label-printer agent for asset tagging

Integration Maturity

Biometric attendance-device connectivity is among the more demanding integration classes in enterprise HRMS deployment, requiring both device-initiated and scheduled-retrieval patterns across heterogeneous customer device estates and network topologies. This capability is operating across 18+ customer environments in production.

12. Summary and Business Value ✅

Technical Position Summary

Dimension	Position
Architecture	Four independently deployable services with workload-class separation over a unified enterprise data platform
Technology	Current-generation .NET 9, React 18 and SQL Server 2022 with no end-of-life runtime dependency
Security	Layered controls across identity, authorization, transport, application data and audit, mapped to the OWASP Secure Coding Practices categories
SaaS	Tenant-partitioned with a three-level organizational hierarchy and commercial packaging model
Scale	Production-validated at 20–41 million rows per core operational table
Integration	Open REST interfaces with proven enterprise system and biometric device connectivity

Business Benefits

Benefit	Basis in the Platform
Reduced implementation risk	Production-proven across 120+ enterprise customers spanning multiple sectors
Lower operational cost	Organizational structure, roles, workflows, policies and templates are configured rather than coded
Independent scaling	Interactive, reporting and batch workloads scale on separate service boundaries
Future-ready platform	Current-generation runtimes across every tier with no end-of-life dependency
Enterprise governance	639 permissioned functions with organizational scope and a complete request-level audit trail
High maintainability	Modular service boundaries reinforced by a structured knowledge model governing every change

Appendix A — Platform Metrics Reference

All figures are measured against the current production implementation.

Category	Metric	Value
Functional	Business domains	19
	Configurable approval request types	40+
	Permission-controlled functions	639
Application	REST endpoints	1,525
	Frontend routes	781
	Frontend modules	26
	Lazy-loaded route boundaries	649
Data	Tables	574
	Columns	9,501
	Schemas	18
	Indexes	699
	Programmable objects	325
	— Stored procedures	235
	— Table-valued functions	25
	— Scalar functions	52
	— Views	6
Reporting	Standard reports	~130
	Dedicated data procedures	~75
Processing	Automated background processes	46
Deployment	Independently deployable services	4
	Backend projects	24
Production Volume	Largest operational table	41.4 million rows
	Core operational table range	20–41 million rows
Adoption	Customers in production	120+
Engineering	Structured knowledge documents	162
	Decision tables	32

Appendix B — Glossary and Acronyms

Term	Expansion	Meaning in This Document
API	Application Programming Interface	Programmatic interface exposing platform capability
CQRS	Command Query Responsibility Segregation	Pattern separating state-changing operations from read operations
DI	Dependency Injection	Composition technique supplying dependencies to a component at construction
ERP	Enterprise Resource Planning	External business system integrated with the platform
HRMS	Human Resource Management System	The platform category
HSTS	HTTP Strict Transport Security	Browser policy enforcing encrypted transport
JWT	JSON Web Token	Signed token carrying identity and organizational-scope claims
OAuth	Open Authorization	Delegated authorization standard used for federated sign-in
ORM	Object-Relational Mapper	Layer mapping database records to application objects
PWA	Progressive Web Application	Installable web application with offline-capable asset delivery
RBAC	Role-Based Access Control	Access model assigning permission sets to roles
RDLC	Report Definition Language Client-side	Fixed-layout report format used for statutory documents
REST	Representational State Transfer	Architectural style of the platform's HTTP interfaces
SaaS	Software as a Service	Multi-tenant delivery model
SPA	Single-Page Application	Frontend architecture serving the full interface from one application shell
SMTP	Simple Mail Transfer Protocol	Protocol used for outbound electronic mail
TVF	Table-Valued Function	Database function returning a result set

Thank You